

Project 1 — Team Proposal (Refreshed, Milestone 1)

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Course: EI 7615 — Deep Learning Architectures

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Project

Discriminative computer-vision pipeline for celebrity identification and detection (Project 1). Milestone 1 establishes a single-face classification baseline over a small CelebA identity subset; Milestones 2-3 extend it to multi-face detection with YOLOv8.

Celebrity identities (final list)

Selected 5 identities from CelebA with 30 images each (a bounded count for balanced classes), chosen to maximize spread across gender, hair color, eyewear, and facial-hair attributes (greedy max-min diversity over `list_attr_celeba.txt`):

Identity ID	# Images	Dominant attributes
6439	30	Blond Hair
5640	30	Male, Gray Hair, Eyeglasses, Bald
6195	30	Male, Black Hair, Mustache
5485	30	Male, Eyeglasses, Wearing Hat
7087	30	Black Hair, Eyeglasses

Framework choice

PyTorch (torchvision for the pretrained ResNet18 backbone). Chosen for: flexible custom `nn.Module` definitions for the from-scratch CNN; torchvision.models gives ImageNet-pretrained weights for the transfer-learning baseline with feature-extraction and fine-tuning modes built in; runs on both CPU (Northeastern Explorer) and any team member's local GPU without code changes.

Division of labor

Task	Owner
Data pipeline, model training (Custom CNN + ResNet18)	Abdellah Faleh
Results write-up (proposal, training-results docs)	Abdellah Faleh
GitHub repo setup & maintenance, reproducibility (README, environment)	Jin-Woo Hong
Evaluation & review (results, confusion matrix/metrics sanity-check, proposal review)	Samuel Tong

Anticipated risks

- **Compute constraints:** Explorer's CPU-only nodes (some without AVX2) can silently kill TensorFlow/PyTorch kernels during model-building ops; mitigated by keeping image size (128x128) and batch size small, and by testing on a different partition/node if a kernel dies with no traceback.
- **Class imbalance:** CelebA identities have uneven image counts; mitigated by selecting identities within a bounded image-count range and stratifying the train/val/test split per identity.
- **Small per-identity sample size:** only 30 images per identity limits how much a from-scratch CNN can learn. This risk materialized during training: an initial custom CNN configuration (heavier augmentation, higher learning rate) collapsed to predicting a single class (20% accuracy, equivalent to random guessing). We mitigated it by lightening augmentation, lowering the learning rate, and reducing dropout, which brought the custom CNN to a genuine 85% test accuracy — still

below the pretrained ResNet18's 95%, but demonstrating real learning. See docs/training_results.md for the full comparison.

- **Milestone 2 dependency:** the model carried forward from Milestone 1 becomes part of the Milestone 2/3 detection pipeline, so architecture/checkpoint format decisions made here should stay compatible with a YOLOv8 integration later.